

“AI-Based Road Accident Prediction: A Machine Learning Framework with Case Study”

A Dissertation

By

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A dissertation Submitted to the Department of Computer Science and Engineering,
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DEDICATION

I wholeheartedly dedicate this project to my beloved parents, whose endless love, encouragement, and sacrifices have been the foundation of my journey. To my respected supervisor and inspiring teachers, thank you for your ongoing support, guidance, and faith in me through every step of this work.

Abstract

Road accidents constitute a serious threat in countries like Bangladesh, often resulting in loss of life, injuries, and significant economic damage. Although various machine learning models such as Decision Trees, Random Forest, and Logistic Regression have been used globally for accident severity prediction, these approaches mainly depend on static datasets and often ignore real-time conditions and the local context. Furthermore, very few studies attempt to explain the predictions, making it difficult for traffic authorities to apply them in real situations. This research aims to address these gaps by proposing a practical framework tailored for Bangladesh. The work reviews and compares earlier models, identifies their shortcomings, and suggests how AI-based prediction systems can be enhanced with real-time inputs and explainability tools like SHAP and LIME. By focusing on accident trends in Bangladesh and showing how data and algorithms can be better aligned with local needs, this study provides a strong base for future road safety applications.

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Declaration of Originality

I certify that except where due acknowledgement has been made, the work is that of the author alone; the work has not been submitted previously in whole or in part to qualify for any other academic award; the content of the thesis is the result of work that has been carried out since the official commencement date of the approved research program; any editorial work, paid or unpaid, carried out by a third party is acknowledged; and ethics procedures and guidelines have been followed.

Author
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Chapter One

Introduction

1.1 Introduction

Road traffic accidents (RTAs) are a critical concern in both developing and developed nations, often resulting in loss of human life, injuries, and considerable financial burdens. In countries like Bangladesh, where road infrastructure, traffic management, and enforcement policies are still developing, road accidents remain a persistent problem. The World Health Organization has reported that low- and middle-income countries account for more than 90% of road traffic deaths globally, despite having only around 60% of the world's vehicles. These alarming statistics call for intelligent, data-driven solutions that can proactively predict and prevent accidents before they occur. Artificial Intelligence (AI) and Machine Learning (ML) have emerged as powerful tools for solving real-world problems, including accident prediction. Traditional methods used for analyzing traffic data and forecasting accidents often rely on static models that do not adapt to new conditions. These models also tend to lack contextual relevance, particularly in localized environments like Bangladesh where road conditions, traffic behavior, and enforcement patterns differ significantly from global norms. Therefore, there is an urgent need to develop a robust, localized framework for predicting road accidents using AI techniques that incorporate real-time data and provide interpretable outcomes for decision-makers. This thesis aims to design and evaluate a machine learning-based predictive model for road accidents that leverages both historical and contextual data. The model will also include explainability features using SHAP (SHapley Additive exPlanations) and LIME (Local Interpretable Model-agnostic Explanations) to ensure transparency and usability in practical deployment.

1.2 Objective

The primary objectives of this research are as follows:

- To develop a machine learning framework capable of predicting road accident risk based on available features such as road type, weather, time, and traffic volume.
- To compare the performance of various machine learning algorithms such as Logistic Regression, Random Forest, Decision Trees, and XGBoost on the chosen dataset.
- To apply model explainability tools such as **SHAP** and **LIME** to interpret and validate the model predictions.
- To customize the framework for the Bangladeshi context by analyzing local accident data (case study).
- To provide a foundation for future deployment in smart city and traffic management systems in Bangladesh.

1.3 Problem Statement

Although multiple studies have attempted to predict road accident severity using machine learning methods such as Decision Trees, Random Forest, and Logistic Regression, most existing approaches still suffer from critical limitations. Previous works, including those by Islam et al. (2023), Ahmed et al. (2022), and Karim et al. (2021), rely heavily on static accident datasets and do not incorporate real-time factors like live traffic conditions, weather, or dynamic road risks. Additionally, these models often ignore local contexts, making them difficult to adapt for Bangladesh, where road infrastructure, traffic behavior, and enforcement differ significantly from other countries. The absence of advanced feature analysis, limited use of explainable AI, and lack of practical deployment strategies further limit their impact on policy and road safety planning. Therefore, there remains a clear research gap in developing a practical, locally relevant conceptual framework that identifies suitable data sources, integrates explainable AI methods, and offers realistic guidelines for applying predictive analytics to road safety in Bangladesh.

1.4 Feasibility Study

The proposed research is practical and achievable within a realistic time frame and available resources. This work will involve a thorough review of existing AI-based accident prediction models, an analysis of their strengths and limitations, and the development of a practical framework adapted to Bangladesh's road safety situation. To demonstrate the viability of the proposed framework, the study will include an experiment using an open-source road accident dataset from sources such as Kaggle. The dataset will be cleaned and prepared, then used to train and test machine learning models such as Random Forest or Decision Tree. The results will show how accident severity can be predicted using real data and provide insights for future real-time applications. From a cost perspective, the project is highly feasible because it does not require any expensive equipment or paid software. All tools including programming libraries and data analysis platforms are open-source and freely available. The research will be carried out within a reasonable time frame that covers literature review, data collection and preparation, model training and testing, framework design, and report writing. Considering the low cost, manageable scope, and useful findings it will generate for future road safety improvements, the project is both realistic and worthwhile.

Chapter Two

METHODOLOGY

2. METHODOLOGY

2.1 Literature Review A significant amount of research has been done in recent years to apply machine learning algorithms for predicting road accident severity. For example, Islam et al. (2023) presented a study using various supervised learning models, including Decision Trees, Random Forest, and Logistic Regression, on an accident dataset to predict the likelihood of severe outcomes. Their research highlighted the potential of ML models to achieve decent accuracy but relied mainly on static, historical data without considering real-time traffic or weather conditions. Similarly, Ahmed et al. (2022) focused on Random Forest techniques for accident prediction and severity analysis using open-source datasets from the UK and India. While their model achieved acceptable prediction accuracy, their approach lacked advanced feature engineering and did not include explainable AI methods such as SHAP or LIME, which are crucial for understanding the factors behind the predictions. Moreover, their framework did not address how such a system could be practically implemented or scaled in the local context of Bangladesh. Karim et al. (2021) explored the use of Decision Trees and Logistic Regression for accident analysis but faced similar limitations. Their model was tested on a structured dataset, but no spatial or temporal hotspot analysis was included, and there was no integration with real-time data sources like live traffic feeds or road condition updates. From these studies, it is clear that while machine learning methods have demonstrated promise, there are key areas that need improvement. The existing works mostly depend on static datasets and fail to address the practical challenges of real-time prediction. The lack of localized case studies for countries like Bangladesh and the minimal use of explainable AI techniques indicate a gap that this proposed research aims to fill by designing a comprehensive framework adapted for local needs.

2.2 Gap Identification

Based on the literature review, several important gaps are clear. Firstly, most existing works rely heavily on static, historical datasets and do not include real-time traffic conditions, weather updates, or live road risk factors. Secondly, advanced AI methods such as ensemble models, deep learning, or hybrid frameworks are rarely explored in the local context. Thirdly, explainable AI techniques like SHAP or LIME are missing or underused, making it difficult for policymakers and traffic authorities to trust and act on the model's predictions. Lastly, there is no practical roadmap for adapting these models to the unique traffic behavior, data limitations, and policy environment in Bangladesh.

2.3 Case Study Review

This research will include a contextual case study review focusing on Bangladesh's road safety scenario. It will examine trends in accident severity, common causes of accidents, and limitations in existing monitoring and reporting systems. Official reports from the BRTA, WHO road safety statistics, and relevant local news sources will be reviewed to understand the unique challenges. This insight will help ensure that the proposed framework is realistic, relevant, and adaptable to local conditions, instead of depending solely on international examples.

2.4 Data Handling and Cleaning

In this research, a publicly available road accident dataset from Kaggle was used to support the initial analysis. Before any form of modeling or interpretation could take place, it was essential to ensure that the dataset was properly cleaned and structured.

2.4.1 Addressing Missing Information

The raw dataset included some incomplete records, particularly in columns such as weather conditions, visibility, and road surface type. To maintain the overall quality of the data, rows with extensive missing values were removed. For features with occasional gaps, standard techniques like using the most frequent value (mode) for categorical attributes and median values for numerical fields were applied to fill in the blanks.

2.4.3 Consistency and Formatting

To ensure uniformity, data types were verified and corrected where necessary. For example, some columns containing numerical values were stored as strings, which were converted to the appropriate format. Categorical fields were also standardized by unifying spelling variations and capitalizations to avoid unintentional splitting during analysis.

2.4.4 Exploratory Assessment

Basic exploratory techniques, including frequency counts and simple plots, were used to gain an initial understanding of how different variables such as time of accident, weather condition, and road type were distributed. These early insights helped identify patterns and ensured that the dataset was ready for any further analytical steps.

Chapter Three

Expected Outcomes

3.1 Expected Outcomes

This research is expected to produce several valuable outcomes. First, it will contribute a clear, well-structured paper showing how artificial intelligence and machine learning models can be applied to predict road accident severity more effectively than traditional methods. Second, the study will include comparative findings based on open-source datasets, highlighting how simple classifiers like Random Forest or Decision Tree perform on real accident data. In addition, the work will deliver a practical conceptual framework that outlines how historical data, real-time inputs, and explainable AI modules can be combined to improve accident prediction. This design will provide a roadmap that local authorities, researchers, or policymakers can adapt in future projects. Finally, the research will generate practical recommendations for collecting real-time data, improving local data availability, and designing systems that can be realistically implemented to support road safety management in Bangladesh.

3.2 Conclusion

This research aims to address the persistent issue of road accidents by proposing a machine learning-based framework specifically adapted to the context of Bangladesh. While previous studies have explored the use of AI in accident prediction, many of them rely solely on historical datasets, overlook real-time factors, and fail to consider the unique challenges present in developing countries like Bangladesh. The proposed work goes beyond existing approaches by incorporating a localized case study, identifying key research gaps, and outlining a practical solution that combines both historical data and real-time variables. The framework also emphasizes the importance of explainable AI, enabling traffic authorities and policymakers to better understand and utilize predictive outcomes.

By conducting experimental validation and reviewing real-world accident trends, this study offers a structured path toward building smarter, more effective road safety systems. It not only contributes to academic knowledge but also holds practical value for future implementations aimed at reducing accident severity and improving public safety.

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